Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Supply Chain Efficiency** "Interoperable Supplier Data Exchange Using SCSN, Digital Product Passports, and TNO Data Ecosystems Solutions"			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility (√), Next-gen Nuclear Power (√), Advanced Bio ( ), Aerospace/Marine (√), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Information & Communication		Knowledge Services	
	Medium Category	Small Category	Medium Category	Small Category
	RFID / USN Services	*Service Platforms and Application Software*	R&D / Engineering Services	Production Management / Simulation / Process Modeling Technologies
Concept of Proposed Technology	Global manufacturing supply chains increasingly depend on fast, secure, standardized data exchange across companies to reduce lead times, eliminate errors, and ensure product lifecycle transparency. However, most SMEs do not have interoperable systems that allow them to share the right data with suppliers and customers, in the right format, at the right time. TNO proposes a joint development program with Korean SMEs to deploy and industrialize: **SCSN (Smart Connected Supplier Network)** • A standardized, secure, peer-to-peer data exchange network originally developed in Dutch high-tech industry for exchanging orders, forecasts, quality data, capacity information, and logistics data between SMEs and OEMs. • It eliminates manual handling, reduces administrative workload, and enhances supply chain visibility. **Digital Product Passports (DPP)** • Standardized, digital product lifecycle records containing manufacturing data, material declarations, traceability			

information, repair/reuse instructions, and environmental indicators.

- DPPs enable compliance with EU regulations and support sustainable circular business models.

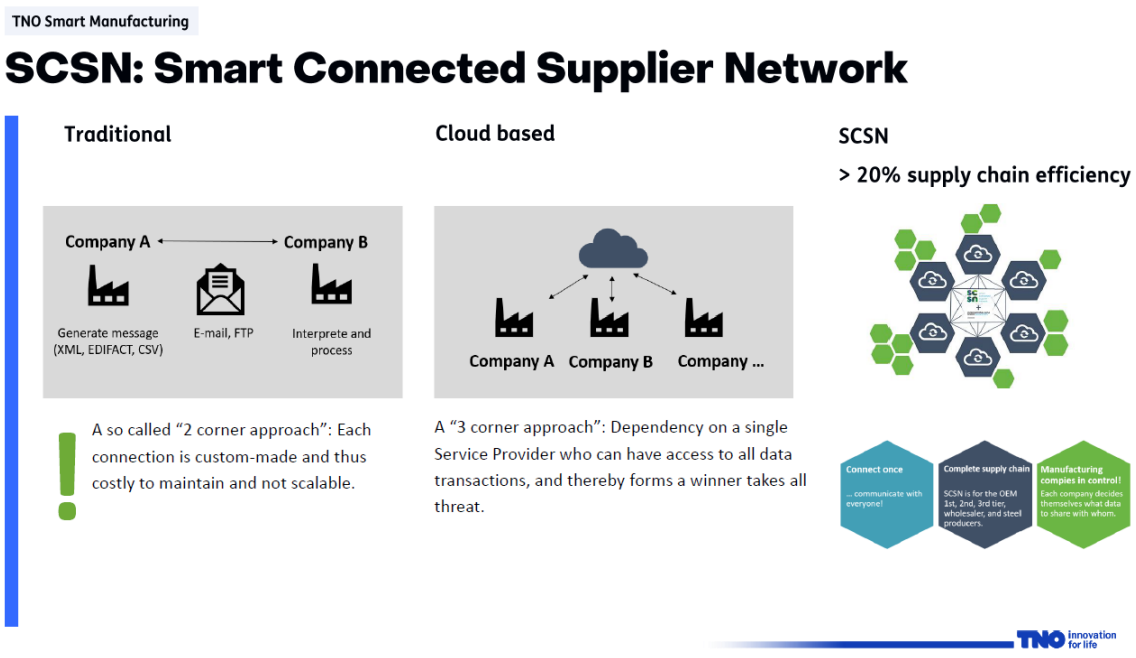
TNO Trusted Data Ecosystems

- A configurable and secure data-sharing architecture enabling companies to exchange sensitive operational data while maintaining data sovereignty, access control, encryption, and auditability.
- These ecosystems allow multi-tier supply chain collaboration without forcing SMEs into a single platform.

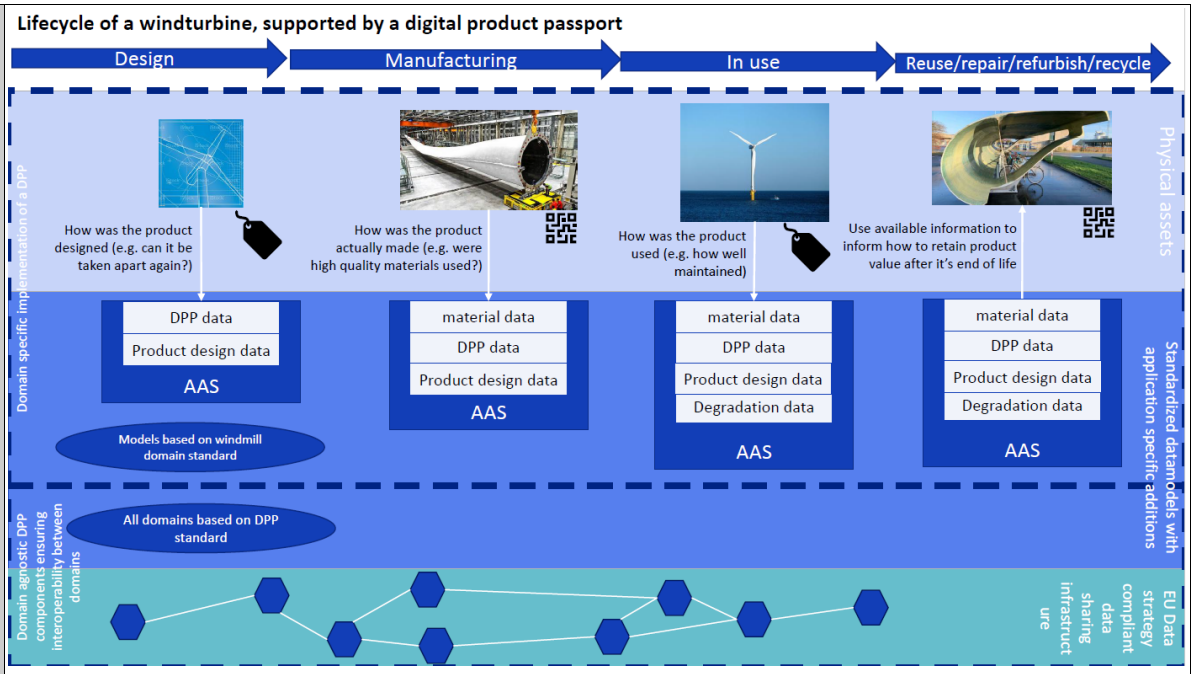
By combining SCSN (transaction data), DPP (lifecycle data), and TNO data ecosystems (governance & trust), SMEs can implement seamless and secure cross-company digital workflows that lead to 20% or more supply chain efficiency improvement, through:

- fewer manual interventions
- reduced lead time variability
- error-free digital order handling
- automated traceability & compliance
- more accurate planning & forecasting
- real-time supplier performance insights

This approach is designed to be low-threshold, incremental, and immediately relevant for semiconductor equipment, robotics, machinery suppliers, automotive components, and precision manufacturing SMEs.



Rationale for Joint R&D Project



A joint project between TNO and Korean SMEs is crucial because:

1. Korea's manufacturing SMEs face rising requirements from global OEMs

- OEMs increasingly demand standardized digital interaction: real-time delivery status, digital quality records, and lifecycle transparency. SCSN and DPP directly address these needs.

2. Data exchange across companies remains fragmented

- Most SMEs operate heterogeneous ERP/MES systems and exchange data through email, portals, or custom integrations.
- SCSN and DPP offer a unified, internationally compatible standard.

3. TNO brings unique expertise

- TNO has built European data ecosystems for high-tech supply chains
- led SCSN development and validation with >200 companies
- contributed to EU DPP technical frameworks
- developed reference architectures for secure data sharing and data sovereignty

4. Korean SMEs bring strong engineering and manufacturing capabilities

Their domain knowledge enables rapid real-world testing and adaptation for:

- semiconductor clusters (equipment, subsystems, tooling)
- automation and robotics suppliers
- mobility and battery value chains
- precision machining and assembly SMEs

5. The combined solution has clear commercial potential

- Korean SMEs can adopt SCSN and DPP technologies early, enabling competitive advantage in the Asian and European markets.

In general it expected SCSN and DPP result in 20-80% supply chain efficiency improvements (depending on current state of integration and digital maturity).

Key Words	1. Smart Connected Supplier Network (SCSN) 2. Digital Product Passport (DPP) 3. Data Ecosystems 4. Interoperability / Industry 4.0 5. Supply Chain Efficiency (20% to 80% improvement)

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	*TRL 4 (Lab validation of integrated modules)*	**End of the Project**	*TRL 6 (Pilot demonstration in SME environment: Production & performance evaluation of prototypes at a pilot scale)*
R&D Objectives	**Key R&D objectives are:** 1. SCSN-compatible connectors for ERP/MES systems used by Korean SMEs. 2. DPP generation & management modules for product lifecycle data. 3. Secure data-sharing architecture using TNO data ecosystems (governance, access control, trust frameworks). 4. Cross-company demonstrators exchanging orders, forecasts, product data, and quality data. 5. KPI-driven supply chain optimization models, targeting 20% efficiency improvement The overarching R&D objective of this joint program is to design, develop and validate an **interoperable, secure, and scalable supplier data-exchange ecosystem** that seamlessly integrates three complementary technologies: **SCSN**, **Digital Product Passports**, and **TNO Trusted Data Ecosystems**. Together, these components will enable Korean SMEs to participate in next-generation international supply networks through standardized digital transactions, trusted lifecycle data sharing, and automated compliance with EU sustainability legislation. A central objective is to **co-create and refine the R&D roadmap in close collaboration with the participating Korean SMEs**. Their concrete operational use cases, related to semiconductor equipment, robotics, mobility components, and precision manufacturing will guide the prioritization of message types, lifecycle data models, interoperability requirements, and SME-specific adoption pathways. During the first phase of the project, each SME will formulate its priority business cases (e.g., automated order processing, quality transparency, spare-parts traceability, predictive maintenance integration), which will be translated into detailed functional R&D requirements. Building on the SCSN model, the project will develop **Korean-localized SCSN connectors** to ensure low-threshold adoption for SMEs with heterogeneous ERP/MES systems. For DPP, the objective is to operationalize **DPP modules** for product genealogy, material declarations, and lifecycle events, ensuring full alignment with the emerging ESPR requirements. Inspired by TNO's **Sm4rtenance** approach, the architecture will incorporate capabilities for **maintenance-related data collection, condition and degradation information, and asset-centric knowledge graphs**, enabling future integration with predictive maintenance and circular business models. Ultimately, the R&D objectives are aimed at achieving actionable improvements in supply chain performance including reduced administrative workload, higher data quality, better planning, and full lifecycle transparency, while enabling Korean SMEs to meet EU market expectations and strengthen their competitive position in global value chains.			

R&D Target Specifications	**Key R&D Target Specifications are:** The joint R&D programme focuses on developing an **interoperable and secure supplier data-exchange solution** that integrates three complementary technologies: 1) **SCSN** for standardized transactional messaging, 2) **Digital Product Passports (DPP)** for lifecycle traceability and compliance, and 3) **TNO Trusted Data Ecosystems** for secure, sovereign cross-company data sharing. R&D target specifications, hyper efficient connected value chain ecosystem: • 20% reduction in administrative handling of orders & confirmations • 15–25% improvement in forecast accuracy across 1–3 tiers • 100% digital traceability via DPP • 30–50% reduction in manual data entry • Secure, standards based integration with at least 3 SME ERP/MES systems • Zero trust architecture for supplier data sharing The target specifications for the joint R&D work focus on the delivery of a fully interoperable, secure, and standards-aligned digital infrastructure, integrating transactional data (SCSN), lifecycle and circularity data (DPP), and trust mechanisms (TNO data ecosystems). These specifications will be further refined and detailed based on the concrete use cases of the Korean SME partners, ensuring that each technical target addresses a real operational bottleneck or opportunity. The target specifications include: **1. SCSN-based transaction interoperability** Implementation of SCSN profiles for orders, order confirmations, forecasts, ASN/transport data, and quality notifications. Localization of message semantics to Korean SME environments while guaranteeing backward compatibility with international SCSN profiles.Connectors enabling plug-and-play integration with at least three different ERP/MES platforms commonly used by Korean SMEs. **2. Digital Product Passport capabilities (CIRPASS-aligned)** Support for default format and alignment with CIRPASS reference architecture (identity, access control, redirection services, lifecycle API). Machine-readable DPP templates based on modular validated ontology structures, supporting model-, batch-, and item-level data as required by the use case.Full representation of product genealogy, bill of materials, maintenance/reuse history, and environmental indicators. **3. Data sovereignty & security architecture** Implementation of a Zero Trust approach for intercompany data exchange, including identity verification, policy enforcement, and end-to-end encryption. A trusted data ecosystem incorporating access control, verifiable credentials, and audit logs, meeting EU expectations for secure cross-company data governance. **4. Integration of Sm4rtenance concepts** Methods for collecting asset degradation data, maintenance events, and reliability parameters.. Ability to link DPP entries to knowledge graphs supporting predictive maintenance and data-driven service models. **5. Quantitative performance targets** • ≥20% reduction in administrative handling time. • ≥30% reduction in manual data entry via automated machine-to-machine data exchange. • ≥15–25% improvement in forecasting accuracy across 1–3 tiers. • 100% digital traceability via DPP compliance. • Full interoperability across at least two multi-tier Korean supply chain use cases.

R&D Plan	These target specifications may be further refined after SME workshops in Korea ensure the solution strongly matches local manufacturing realities. The R&D work proceeds through four integrated streams: 1. **Architecture & Standards Alignment (M1–M6):** Define functional requirements, message profiles, DPP schema, security model and reference architecture combining SCSN, DPP and the data-ecosystem components. 2. **Technology Development (M4–M18)**: Develop ERP/MES connectors, DPP services, and security components. Implement message translation, product genealogy, zero-trust access control and audit logging. 3. **Integration & Pilot Demonstration (M10–M24):** Integrate the components into two multi-tier demonstrators. Execute field trials with real orders, forecasts and lifecycle data exchanged across Korean and Dutch SMEs. 4. **KPI Measurement, Optimization & Validation (M6–M26)**: Measure improvements in administrative workload, planning accuracy, traceability and data quality. Validate the targeted 20% efficiency gain and produce implementation guidelines for wider SME adoption.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	**Key R&R TNO:** • Share and further develop SCSN platform • Stronger Korea–Netherlands cooperation on interoperable supply chain technology • Commercial ready SCSN connectors for Korean SMEs • DPP generation tools compatible with emerging EU regulations • Increased competitiveness through 20% supply chain efficiency gains • New digital services and business models based on real time supplier data • Enhanced reputation and export opportunities for participating SMEs *TNO's primary responsibility is to provide the technical expertise, reference architectures, and development frameworks for interoperable supply chain data exchange. This includes the further development and internationalization of SCSN, translation of CIRPASS DPP architecture into implementable modules for Korean SMEs, and the setup of a trusted data ecosystem based on TNO's secure data sharing technologies.* *TNO will lead the research on semantic interoperability, security design, and system architecture, ensuring alignment with European regulatory developments (ESPR, DPP delegated acts). TNO will also guide the integration of Sm4rtenance like maintenance data concepts to support predictive maintenance and circular economy use cases.* *A core responsibility of TNO is to co create the detailed R&D objectives together with the Korean SME partners, ensuring the technology development is tightly aligned with real operational needs and accelerates SME adoption. TNO will conduct the cross-company pilot design, KPI framework creation, and validation of performance improvements.*
Expected R&R of Korean SME Partner	Key R&R Korean SMEs: • Provide factory environments, subject-matter experts, and operational data for localization and testing. • Lead integration into Korean MES/QMS platforms and support UI/UX localization. • Conduct on-site trials, gather operator feedback, measure KPIs, and support refinement. • Prepare commercialization and service models for SCSN and DPP within the Korean market.

	Korean SMEs will act as industrial innovation partners providing real-world environments, domain expertise, and operational data required for meaningful technology development. Their responsibilities include defining their specific use cases (e.g., quality data exchange, configuration management, maintenance records, environmental data capture), which will then be used to refine the R&D objectives and target specifications. The SMEs will lead on ERP/MES integration, localization of UI/UX, execution of field tests, and evaluation of technical and organizational feasibility. They will validate the solution in real production conditions and provide structured feedback for iterative improvement. Korean SMEs will also prepare commercialization models for the Korean market, including service offerings, maintenance of connectors, and potential integration with Korean data space initiatives. Together, TNO and the Korean SMEs will jointly manage the pilot implementation, interoperability tests, KPI measurement, and preparation of exploitation pathways for long-term adoption.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Key impact:** • Commercial-ready **SCSN connectors** for Korean ERP/MES systems • **DPP tooling** compatible with EU regulatory developments • **Zero-trust data-sharing architecture** enabling secure multi-tier exchange • Full **cross-company demonstrators** for two industrial sectors • Verified **efficiency improvements** of ≥20% for participating SMEs • Implementation guides and exploitation plans for broader market uptake The project is expected to generate significant technological, economic, and strategic impact for both Korea and the Netherlands. The outcomes include commercially ready SCSN connectors, CIRPASS-aligned Digital Product Passport tooling, and a trusted, secure cross-company data-exchange architecture that can be deployed by Korean SMEs in multiple industrial sectors. By aligning SCSN, DPP, and trusted data ecosystems, the project will enable Korean SMEs to meet the increasing international demand for digital transparency, regulatory compliance, lifecycle reporting, and high-performance supply chain collaboration. The integration of Sm4rtenance principles will further enable smarter maintenance, improved asset uptime, and enhanced circularity pathways. From a broader perspective, the project will strengthen Korea–Netherlands industrial cooperation, positioning both ecosystems at the forefront of interoperable global supply networks. Korean SMEs will benefit from reduced manual workload, improved planning and forecasting, automated compliance reporting, and the ability to participate in advanced European manufacturing ecosystems such as semiconductors, robotics, machinery, and mobility systems. Importantly, the impact will be further maximized by tailoring each solution component to the specific use cases identified by the participating Korean SMEs during the early phase of the project. This ensures the resulting technologies are directly applicable, economically viable, and scalable across the Korean manufacturing landscape.

6. References

TNO SCSN: Value chain integration | TNO